

A160 – Quantum Mechanics on T^4 : Bell as Topology

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.1 The Thesis

[STIPULATION] Bell correlations are on T^4 a *topological* phenomenon: two measurement sites that lie far apart in the unrolled picture are connected in the compact picture by a closed path.

What appears as action at a distance in the projected picture is proximity in the compact picture. Non-locality is thereby an artefact of the projection (A090), not an ontological property of the world.

[B] This is not a circumvention of Bell's theorem. Bell rules out local hidden variables *in the projected space*. A topological connection in the pre-ordered space is not a hidden variable in the sense of the theorem – it is not an additional quantity that fixes the outcome, but a statement about the geometry in which both measurement sites lie.

.2 What Does *Not* Change

[K] The predictions of quantum mechanics remain. The CHSH value for the singlet state is

$$S = 2\sqrt{2} = 2.8284271 \dots$$

and the Tsirelson bound holds unchanged.

[B] In particular, the mass extension changes nothing: the $\mathbb{C}^3$ tensor factor is invisible for correlation quantities, because the fibre state is normalised and the observable acts there as the identity (A070). This is computed, not claimed.

A theory that changed something here would already be refuted – CHSH is one of the best-tested quantities in physics.

.3 The Residual of Order ξ

[K] Compactness has not completely disappeared after the projection. It leaves a residual, and this residual is the only place where the theory can show itself in the quantum domain at all.

The order of magnitude is fixed by ξ : the relative deviation is about 10^{-4} to 10^{-5} , depending on how many ξ -factors enter.

What this means for measurability – without gloss. The effect is *not resolvable* with current hardware. The systematic uncertainties of real Bell experiments – detector efficiency, angle calibration, state preparation – are orders of magnitude larger. A fit of the ξ parameter to measured CHSH values is therefore *pointless*: it would adjust hardware systematics, not measure geometry.

This finding is explicitly part of the result. A prediction whose signal lies below the noise is a prediction for later, not a confirmation for today.

.4 Why the Residual Is Nevertheless a Prediction Quantity

[K] The ξ -order of the CHSH deviation is one of the four prediction quantities of the A-series (A010). The reason is not that it is testable today, but that it is *in principle* testable and the theory can fail at it.

Three properties make it a genuine prediction:

- The *direction* is fixed: the residual reduces the correlation, it does not increase it.
- The *order of magnitude* is fixed and not adjustable – it depends solely on ξ , which is stipulated in A020 and not readjustable.
- The *scale dependence* is fixed: the residual grows logarithmically with the number of degrees of freedom involved, not linearly.

[STIPULATION] If a CHSH value is measured that *exceeds* the Tsirelson bound, or one that falls below it by significantly more than ξ -order without hardware systematics explaining it, the theory is refuted at this point.

.5 The Geometric Reading of Entanglement

[B] In the compact picture, an entangled pair is not a pair of separated objects with a connection but *one* object with two projection images. The correlation is then not a transmission but identity.

From this it follows immediately why no information is transmitted and Relativity remains unviolated: there is nothing that travels from A to B. Signal-freedom is in this reading not an additional condition to be imposed, but a consequence of the construction.

[S] The assignment of which closed paths correspond to which concrete entanglement states has not been worked out. It is a sketch and is marked as such.

.6 Verification Script

- CHSH in the singlet vs. the Tsirelson bound, the invisibility of the $\mathbb{C}^3$ factor, and the order of magnitude of the ξ -residual compared to typical experimental uncertainties: `python/A_Serie_Skripte/a160_chsh.py`

.7 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
S_{CHSH}	CHSH correlation value; classical ≤ 2 , Tsirelson $\leq 2\sqrt{2}$
Δ_{CHSH}	Deviation from Tsirelson value; T0 prediction $\sim \xi$
ϵ_{CHSH}	Relative CHSH correction from T^4 geometry

.8 Open Questions

First, the prefactor $1/(2\pi)$ is now geometrically justified. ξ is the fractal correction to the arc length on the torus. A CHSH measurement is an angle measurement over a full circle (circumference 2π). The correction per angle unit is $\xi/(2\pi)$ – the circumference is the conversion factor, not a free parameter. **[B]**

[B] Connection to the cumulative formula (A165). The elementary formula $\xi/(2\pi)$ and the cumulative formula $\text{CHSH}^{\text{T0}}(n) = 2\sqrt{2} \cdot \exp(-\xi \ln n / D_f)$ describe different observables:

- **Elementary:** phase correction per single measurement (one circumference 2π) – hence $\xi/(2\pi)$.
- **Cumulative:** damping over $\ln n$ entanglement layers (logarithmic entropy scale) – hence $\xi \ln n / D_f$. The 2π does not appear because the relevant integration length is $\ln n$, not the circumference.

Why D_f in the denominator? The fractal correction ξ acts in $D_f = 3 - \xi$ effective dimensions simultaneously. Per dimension: ξ/D_f . The cumulative formula integrates over all dimensions – that is fractal self-consistency: ξ/D_f instead of $\xi/3$.

Both formulas are now geometrically justified. What remains: the measurement. The deviation lies below the NISQ noise (A165).

Second, the path assignment is open (see above, **[S]**).

Third, the claim that Bell is thereby ontologically defused is an interpretation. It is consistent, but it does not follow compellingly – the same mathematics can be read without this interpretation. The A-series does not claim to have decided the interpretational question.

.9 Supersedes

This document subsumes: Dok. 020–023 (quantum mechanics, Bell), Dok. 035 (non-locality), Dok. 073 (entanglement), Dok. 142 (CHSH), Dok. 055 (dynamic mass photons, energy-dependent non-locality), Dok. 074 (no-go theorems and T0 answer: Bell’s theorem, Kochen–Specker, locality argument), Dok. 131 (apparent instantaneity, local vs. global relations, Green’s function and causality in the T0 formalism). Incorporated: R41.

.10 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.